INCH-POUND

MIL-M-38510/13F
AMENDMENT 6
25 August 1993
SUPERSEDING
AMENDMENT 5
31 May 1985

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, TTL,
COUNTERS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/13F, dated 16 April 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Delete and substitute as printed above.

1.1, delete and substitute:

"1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, binary and decade counters. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number."

1.2.3, delete letter B in its entirety and substitute:

"B F-3 (14-lead, 3/16" x 1/4"), flat package".

PAGE 2

1.3, delete "Thermal resistance, junction-to-case (Θ_{JC}) 0.09°C/mW for flat package; 0.08°C/mW for dual-in-line package" and substitute:

"Thermal resistance, junction-to-case (Θ_{JC})
Cases A, B, C, D, E, and F - - - - - - - - - - - - - (See MIL-M-38510, appendix C)."

1.3, after junction temperature (T_J): Add "2/".

1.4, after supply voltage: Add "(V_{CC})".

1.4, width of clock pulse, $t_{w(CLOCK)}$, device types 03, 04, 05, and 06: Delete "30 ns" and substitute "25 ns".

1.4, setup time, $t_{(SETUP)}$, device types 03, 04, 05, and 06: Delete "Enable P - - - - - - 25 ns minimum".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
41	41
42	42

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1.4, following hold time operating conditions, add enable time conditions as follows:

"Enable time, $t_{(ENABLE)}$, device types 03, 04, 05, and 06, Enable P - - - - - - - - - - 25 ns minimum".

Following footnote 1/, add:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 3

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

3.6, delete and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

PAGE 4

* TABLE I, I_{IL2}, device types 08 and 09, minimum limits column: Delete "-0.7" and substitute "-0.5".

TABLE I, I_{IL5}, device types 08 and 09, minimum limits column: Delete "-0.7" and substitute "-0.5".

* TABLE I, I_{IL6}, device types 08 and 09, minimum limits column: Delete "-0.7" and substitute "-0.5".

PAGE 6

* TABLE II, group A test requirements, class B devices column: Add "8".

PAGE 7

* 4.2a(1), delete and substitute as follows:

"(1) Test condition D or E, using the circuit shown on figure 5, or equivalent. Test condition A and the applicable test circuit shall be allowed with the approval of the qualifying activity."

4.2c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

* 4.4.3c(1), delete and substitute as follows:

"(1) Test condition D or E, using the circuit shown on figure 5, or equivalent. Test condition A and the applicable test circuit shall be allowed with the approval of the qualifying activity."

* 4.4.3c(3), delete "appendix B of MIL-M-38510" and substitute " method 1005 of MIL-STD-883".

PAGE 8

6.1: Delete in its entirety.

6.4, delete "GND - - - - - - - Electrical ground (common terminal)" and substitute:

"GND - - - - - - - - - - - Ground zero voltage potential".

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PAGES 38 AND 39

FIGURES 6 and 7, note 1, delete "PRR = 1 MHz" and substitute "PRR ≤ 1 MHz".

PAGE 43

FIGURE 8C, note 1, delete "t_p = 20 ns" and substitute "t_p = 25 ns".

FIGURE 8C, note 1, delete "V_{GEN} = 4.5 V" and substitute "V_{GEN} = 3.0 V".

PAGES 44, 45, AND 46

FIGURES 9, 10, and 10A, note 1, delete "PRR = 1 MHz" and substitute "PRR ≤ 1 MHz".

PAGE 49

TABLE III, footnote 14/, delete and substitute as follows:

"14/ The limits shown shall be as follows:

Test	Min/Max limits (mA) for circuit				
	A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.4/-1.3	-0.3/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-1.4/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-1.4/-4.8	-0.7/-4.8	-0.7/-4.8	-1.4/-6.4	-1.4/-6.4

"

PAGE 52

TABLE III, footnote 14/, delete and substitute as follows:

"14/ The limits shown shall be as follows:

Test	Min/Max limits (mA) for circuit				
	A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.4/-1.3	-0.4/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2

"

PAGE 53.

TABLE III, test numbers 43 and 45, minimum limits column: Add "9/" after -0.7. Make this same change on pages 58 and 68.

PAGE 57

TABLE III, add new footnote 9/ as follows:

"9/ Minimum limit for circuit C shall be -0.5 mA." Make this same change on pages 62, 67, and 72.

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PAGE 63

TABLE III, test numbers 43 through 45, minimum limits column: Add "9/" after -0.7.

PAGE 64

* TABLE III, test numbers 47 through 51, pin 2: Delete "D".

PAGE 72

TABLE III, subgroup column: Add "11"; conditions: Add "Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C."

PAGES 80 AND 86

TABLE III, device types 08 and 09, subgroup column: Add "8"; conditions: Add "Repeat subgroup 7 at T_C = +125°C and T_C = -55°C."

PAGE 82

* TABLE III, device type 09, test numbers 1 through 4, minimum and maximum limits columns: Add "19/" after -0.7 and -1.6, respectively.

TABLE III, test number 6, symbol column: Delete "I_{IL8}".

PAGES 85 AND 86

TABLE III, subgroup column: Delete "T_C = 125°C" and substitute T_C = 25°C".

PAGE 87

TABLE III, footnote 4/, delete and substitute: "See figure 10A. Apply waveform E for tests 161 and 178 and waveform H for tests 162 and 179."

* TABLE III, add footnote 19/ at bottom of page as follows:

"19/ The limits shall be -0.5 mA minimum and -1.3 mA maximum for circuit E."

PAGE 88

TABLE IV, circuit B column: Add manufacturers designation for device type 09.

The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

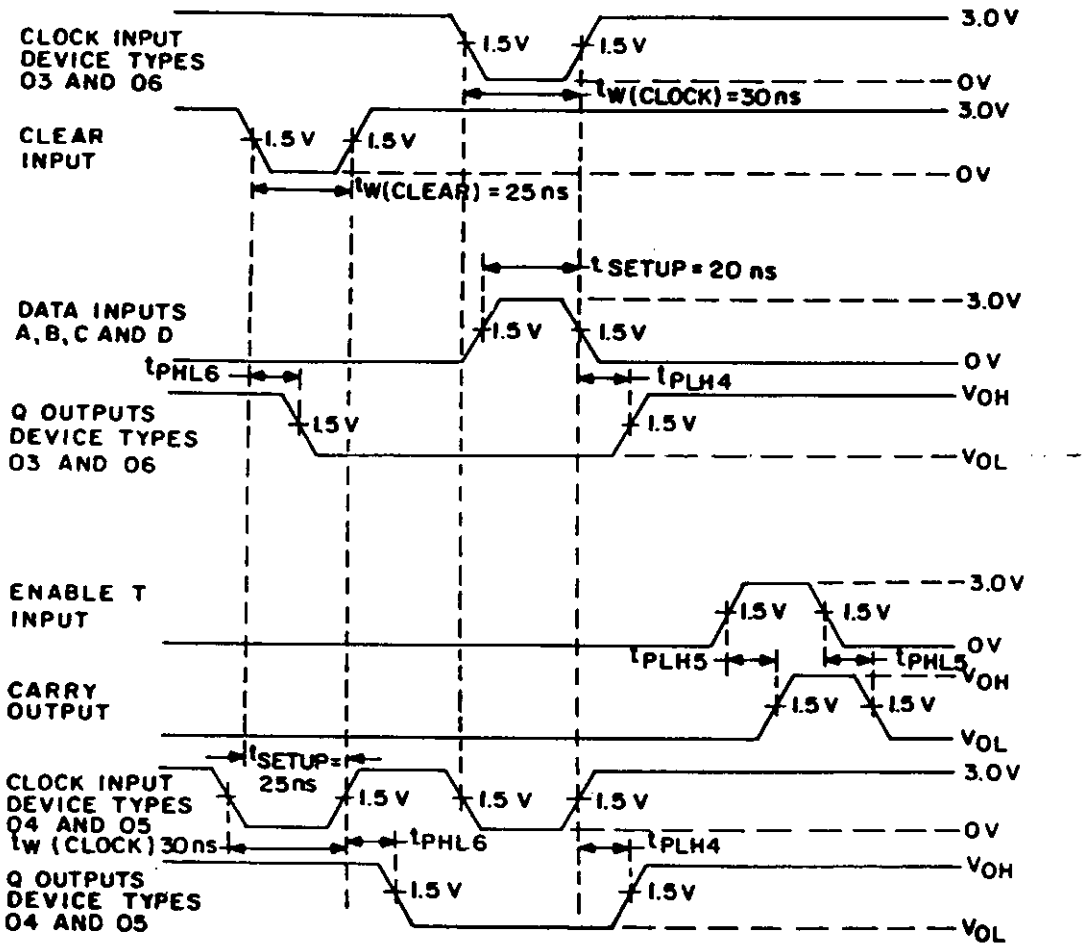
User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1331)

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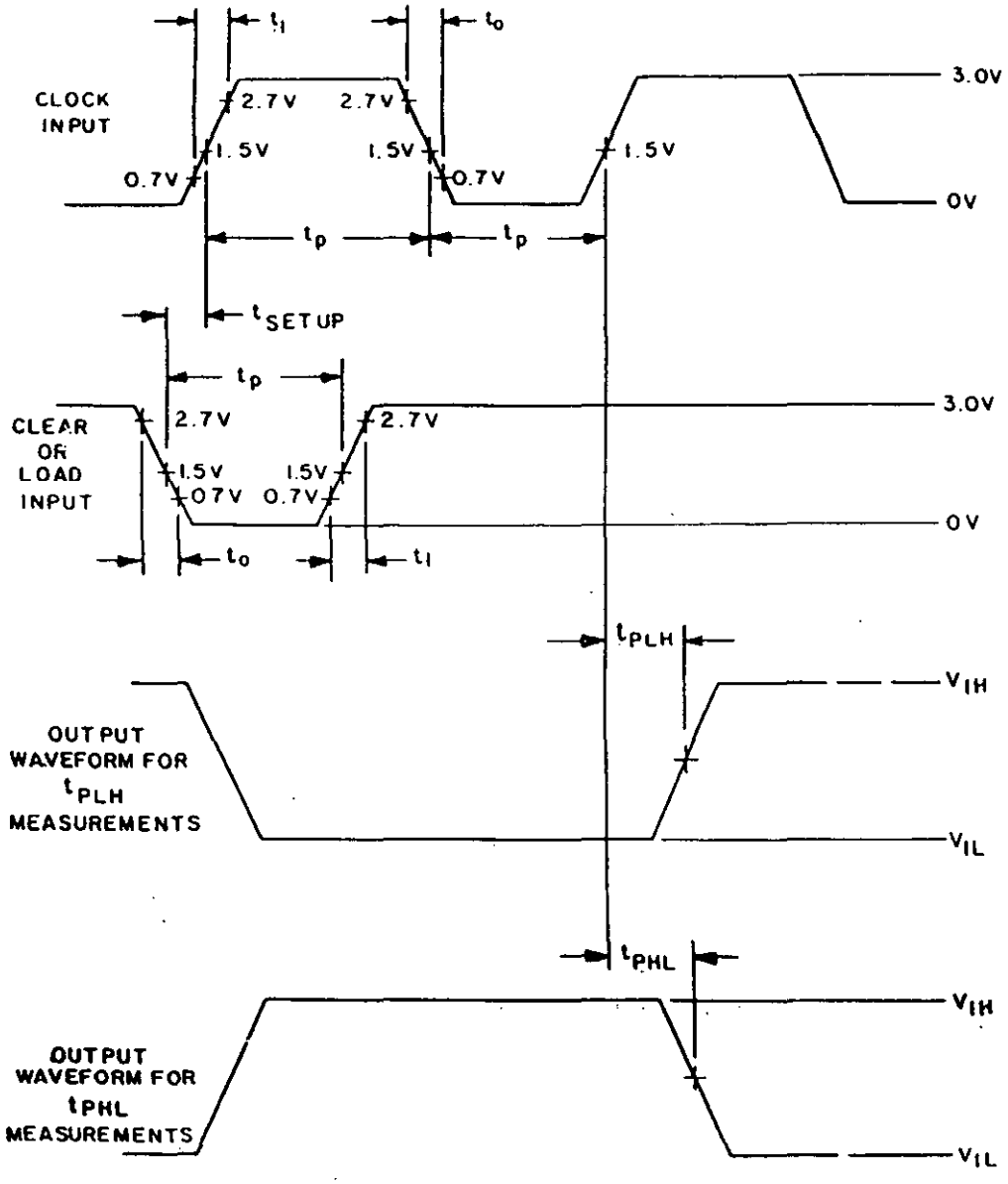


VOLTAGE WAVEFORMS

NOTES:
1. The input pulses are supplied by a generator having the following characteristics: $t_r \leq 10$ ns, $t_f \leq 10$ ns, PRR ≤ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. Enable T duty cycle ≤ 50 %, PRR = 1 MHz.

FIGURE 8A. Switching time test circuits and waveforms for device types 03, 04, 05, and 06 - Continued.

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NOTES:
1. The input pulses are supplied by a generator having the following characteristics: $t_0 = t_1 \leq 10$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. The t_{SETUP} for load pulse = 30 ns; t_{SETUP} for clear pulse = 25 ns.

FIGURE 8B. Switching time test circuits and waveforms for device types 03, 04, 05, and 06 - Continued.

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